SpaceX's Agile Hardware Revolution: Transforming Aerospace Through Software Development Principles

INFO 7374 - Advanced Business Process Engineering II

Name: Sachin Temgar

Date: December 6, 2025

Executive Summary

SpaceX redefined the approach to aerospace development through the systematic implementation of the Agile, Lean, and DevOps concepts to hardware development (Berger, 2020; Weinzierl and Sarang, 2020). Through shortening the development cycles with an iterative approach (Beck et al., 2001), removing wastes with value stream optimization (Womack and Jones, 2003), and adopting continuous integration (Kim et al., 2016), SpaceX managed to decrease the development cycles 2.6 times, cut costs by 90 percent, and compel industry transformation. This discussion indicates that the concepts that find their origin in the world of software development can be given power when applied to the hardware with the help of proper organization structure, corporate values, and technical architecture.

1. Introduction

At the beginning of 2000s, there was a paradox of aerospace development, the more advanced technology was, the longer the timeline and higher the costs were. The Space Launch System developed by NASA took 15 years and cost over 25+ billion with 1970s technology known to be reliable (GAO, 2023). This is the so-called dilemma of the innovator described by Christensen (1997) that is the established organizations caught in the rut of former successful processes. In the meantime, the commercial space opportunity required a radical cost-cutting, which the traditional contractors, who were confined to waterfall processes (NASA, 2018), could not provide.

This was a software problem for SpaceX, which was established in 2002. The mandate of Musk: create rockets like software, iterate, fail fast, learn never stop (Berger, 2020, p. 87). This was in line with Agile principles that focused on an iterative development, customer cooperation, and change response (Beck et al., 2001). In 2025, SpaceX controlled more than half of the commercial launch (50%+ market share) (Sheetz, 2023), which is order-of-magnitude better.

The following analysis is based on the frameworks of Weeks 10-13: Agile principles (Beck et al., 2001; Schwaber and Sutherland, 2020), Lean thinking (Ohno, 1988; Womack and Jones, 2003),

DevOps practices (Kim et al., 2016; Forsgren et al., 2018), and the organizational transformation theory (Kotter, 1996; Denning, 2018).

2. Current State Assessment: Traditional Aerospace Waterfall Model

2.1 The Sequential Development Process and Its Historical Context

Traditional aerospace is based on the Systems Engineering V-Model, a very linear, stage based process that was created in the 1960s (NASA, 2018). This approach was developed in the context of the Cold War when the Apollo program was carried out in rather special circumstances that justified cost-inefficiency. This was due to the perceived existential threat posed on the American technological superiority as a result of the launching of a Soviet satellite Sputnik in 1957 (Launius, 2003). The 1961 promise to land on the moon by President Kennedy before this decade was out gave space exploration a geopolitical need which failure would not only lead to loss of the mission but also humiliation to the nation (Kranz, 2000).

Apollo had access to unmatched resources that would not be available in the present day: the NASA budget reached as high as 4.4% of federal expenditure in 1966 (5.2 billion a year today, or about 50 billion today), as compared to 0.5% (NASA, 2020). The inefficiencies of the waterfall model were possible due to this unlimited funding climate of a staggering 25-billion dollars 1960s or 280 plus billion in today's value (Launius, 2003) which allowed the model to run wild. With 400,000+ workers, mobilized as national priority in government and contractors, undue and redundant trials were cost-effective since speed-to-moon was of more importance than cost (Logsdon, 2010).

Everything changed after the Cold War ended in 1991. The geopolitical imperative disappeared, budgets were reduced to minimal amounts, and commercial market demands substituted the needs of the government (Weinzierl et al., 2018). The approach which had worked successfully in unconcerned times of cost proved unsustainable in urgent, efficiency-seeking times. The processes optimized in one scenario become liabilities as the circumstances vary as pointed out by Christensen (1997). Conventional aerospace operated with waterfall methods of the Apollo program with no Apollo program funds, establishing the unsustainable cost curve which SpaceX disrupted.

The standard 5-10 years cycle consists of: Requirements Phase (12-24 months) freezing specifications on a contract basis; Design Reviews (24-36 months) of thousands of pages of documentation; Manufacturing (24-36 months) by geographically dispersant subcontractors; Integration & Testing (12-24 months) which is catastrophically costly; and First Flight which must work under the mindset of failure is not an option (Kranz, 2000).

2.2 Value Stream Analysis: The Waste Problem

Applying Lean value stream mapping to aerospace development reveals only 28% of time adds value (NASA Vision 2040 Study, 2018):

Process Step	How Long It Takes	Actual Useful Time	What's Slowing Us Down
Requirements	18 months	About 2 months	Waiting around for approvals, excessive documentation
Design reviews	24 months	Roughly 6 months	Stuck waiting for review boards
Procurement	12 months	Maybe 1 month	Contract negotiations drag on, shipping logistics
Manufacturing	30 months	Around 12 months	Building in batches, can't start next step until previous one is done
Integration/Testing	18 months	8 months	Finding problems too late, going back to fix things
Total	**102 months**	**29 months**	**72% of time is wasted**

The Seven Wastes dominate: waiting for approvals, coordinating across organizational boundaries, transporting components nationally, maintaining inventory, discovering defects late, producing documentation for review processes, and reworking designs after integration failures.

2.3 Organizational and Economic Barriers

Matrix Structure: The traditional contractors operate under functional silos (Propulsion Dept, Structures Dept, Avionics Dept) where the matrix management generates a coordination overhead and unclear accountability.

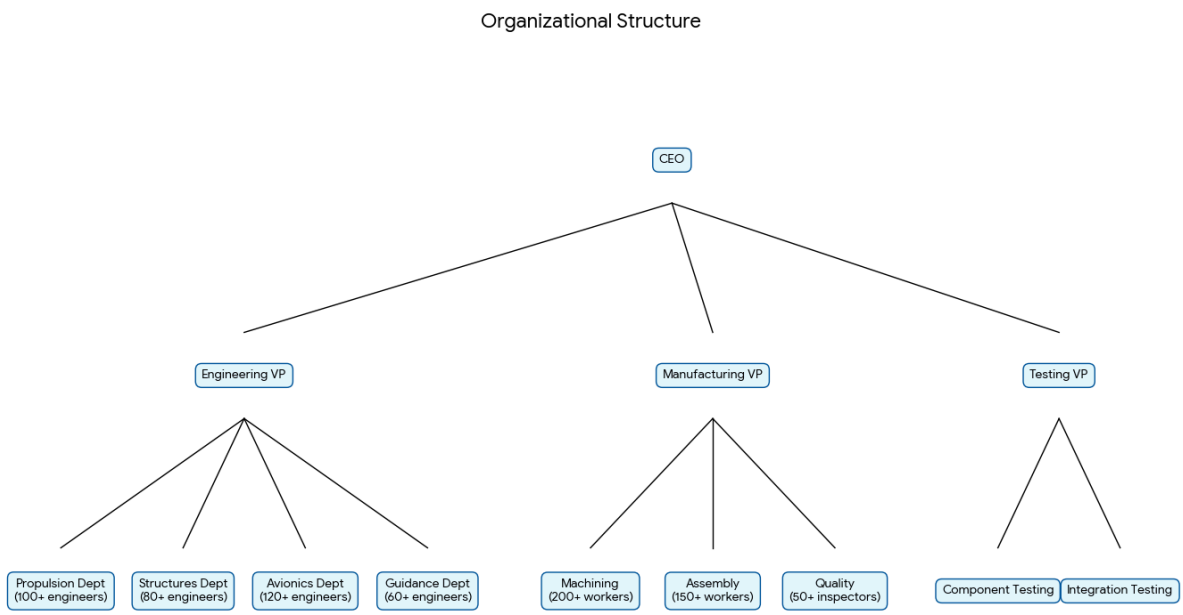


Figure 1. Traditional Aerospace Functional Silo Organization Organizational structure based on NASA (2018) and Galbraith (2009). Employee counts represent typical large aerospace contractor scale.

Risk-Averse Culture: The success of any program in career advancement is measured by success, with no failures, which encourages over-engineering and over-testing instead of innovation.

Cost-Plus Contracting: Government pays all expenses plus profit guarantee (7-15%), which causes perverse incentives, thus the longer duration and the larger the scope the higher the revenue. The case of Boeing Starliner is no exception: 5.7 billion expended and 1.5B + in excess and it is not operational yet (Foust, 2024).

3. Transformation Approach: Applying Agile, Lean, and DevOps to Hardware

3.1 Core Philosophy: Hardware as Iteration Medium

SpaceX's core insight: there is a degree of difference, not a type of difference between the hardware and the software. Speed of iteration is constrained by physical factors, but the same principles of rapid feedback and continuous learning can be used.

Musk's Principles (Berger, 2020):

1. "The best part is no part" (eliminate waste)

2. "Simplify, then optimize" (working solution before perfection)
3. "Accelerate cycle time" (speed as advantage)
4. "Automate" (remove manual bottlenecks)
5. "Fail fast, fail cheap, fail in the open" (learning over perfection)

3.2 Agile Principles in Hardware

3.2.1 Iterative Development Through Prototyping

Starship Program (2019-2024): SpaceX built 15+ prototypes in 5 years, each testing specific hypotheses[1]:

Prototype	Build Time	Test Objective	Outcome	Learning
SN1	4 months	Pressure vessel	Structural failure	Redesigned welds
SN8	5 months	High-altitude flight	Landing explosion	Fuel pressure fix
SN9	6 months	SN8 improvements	Landing explosion	Engine redundancy
SN10	6 months	Multiple fixes	Successful landing	Leg redesign
SN15	6 months	Reliability	Full success	Proven baseline

Each prototype functioned like a software sprint: clear goal, 4-6 month duration, public demo (test flight), retrospective (failure analysis), and immediate incorporation into the next build already under construction. Traditional aerospace would spend 3-5 years simulating these scenarios before building one test article.

3.3 Lean Principles: Waste Elimination

3.3.1 Vertical Integration Strategy

SpaceX doesn't waste money on a traditional supply chain since SpaceX manufactures 80 percent of parts internally:

Traditional Problems:

- Modifications in designs necessitate the adjustments of supplier contracts (months).
- The geographic dispersion forms delays in communication.

- Supplier margins add 20 - 40% cost
- The barriers of intellectual property are no sharing of design.

SpaceX Benefits:

- Changes in design made within the day.
- All teams colocated
- Direct cost control
- Complete design freedom

Impact: The iterations of the Merlin engine have moved 12-18 months of 15-20M (traditional supplier model) to 3 - 6 months of an approximate 1M (in-house), so that the cycles of learning can be created and developed under Agile.

3.3.2 Continuous Improvement (Kaizen)

Kaizen: Each flight instigates instantaneous enhancement cycles (Liker, 2004). Following the explosion at the ground test of Crew Dragon in April 2019, the root cause was found in 2 weeks, redesign in 6 weeks, and in-flight abort success occurred 9 months later- compared to the traditional 12-18 months (Berger, 2020).

Every flight triggers immediate improvement cycles:

Crew Dragon Abort Anomaly (April 2019):

- Capsule exploded during ground test
- Traditional timeline: 12 - 18 months investigation and requalification
- **SpaceX kaizen**:
    - 1) Root cause in 2 weeks
    - 2) Redesign in 6 weeks Testing in 4 months
    - 3) Successful in - flight abort 9 months later

3.4 DevOps in Hardware Manufacturing

3.4.1 "You Build It, You Run It" Ownership

End-to-End Ownership: SpaceX has done away with conventional handoffs (Kim et al., 2016). Every engineer has elements of conception up to flight operation. Raptor engine chief engineer is present on tests and on-the-fly during firings, watching telemetry and issuing redesign directions - establishing direct feedback connections between design choices and operating results.

3.4.2 Continuous Integration and Testing

CI/CD Adapted to Hardware:

- Daily builds: The factory is running 24 hours a day; components are made on a daily basis.
- **Parallel assembly**: 4 - 5 Starship prototypes in various stages simultaneously
- **Test hierarchy**:
    - 1) Every engine test-fired (unit test)
    - 2) Vehicle static fires (integration)
    - 3) Flight tests (system)

Deployment Frequency:

- Falcon 9: 96 launches in 2023 = every 3.8 days
- Individual boosters: B1058 flew 19 times in 3 years
- Traditional: 1 - 2 flights per vehicle per year
- **Improvement**: 10-15x deployment frequency

3.4.3 Observability and Public Transparency

SpaceX instruments every test extensively (thousands of sensors, high-speed cameras, real-time telemetry) and **livestreams publicly, including failures**. This radical transparency:

- Creates external accountability
- Demonstrates cutting-edge work (recruiting)
- Builds public support through excitement
- Makes internal psychological safety credible

4. Implementation Analysis: Organizational Transformation

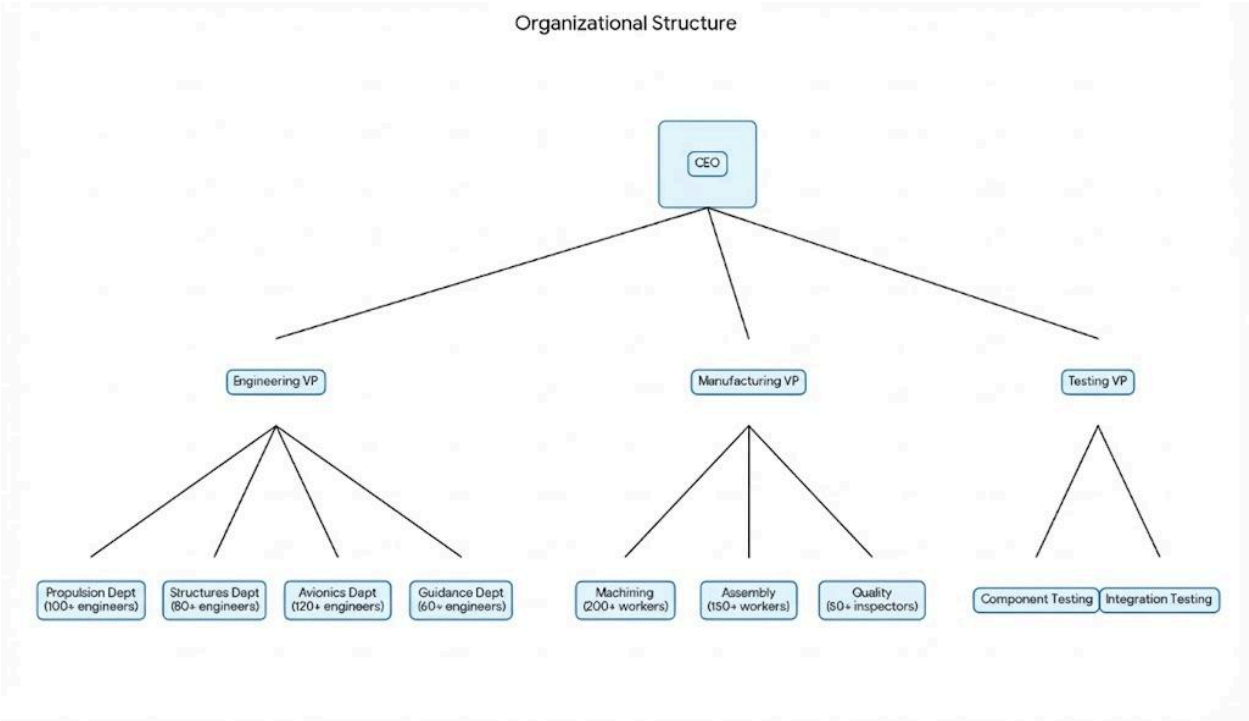
SpaceX rejected functional silos for cross-functional product teams (Denning, 2018). Traditional aerospace separates Engineering, Manufacturing, and Testing departments. SpaceX organized around products: Falcon Program, Starship Program, Starlink Program each with cross-functional teams owning entire value streams (Berger, 2020). This implements Schwaber and Sutherland's (2020) principle that "Scrum Teams are cross-functional...and self-managing" (p. 5).

Teams possess design, build, and test authority without multi-level approvals. Constraints (safety, interfaces, budgets) replace controls. When the Raptor team identified simplification opportunities, traditional aerospace would require 12-18 months of review boards; the SpaceX team presented to Musk in weekly review, received approval at the same meeting, implemented in 6 months (Berger, 2020).

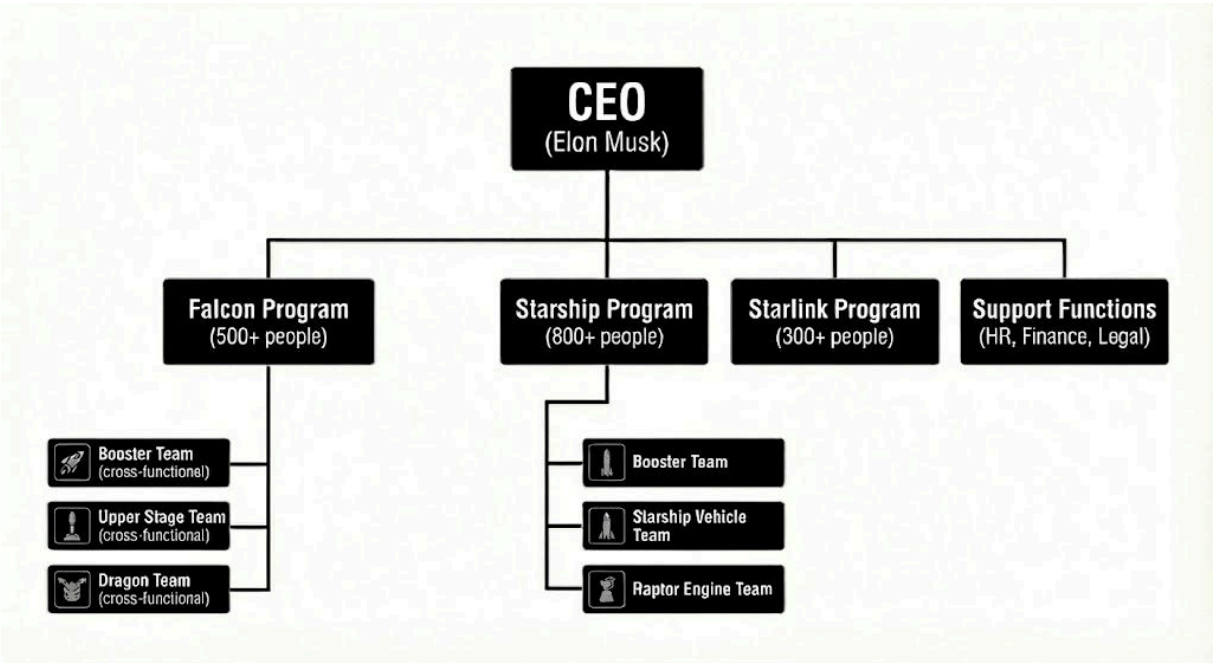
4.1 Structural Transformation

4.1.1 From Functional Silos to Cross-Functional Teams

Traditional Matrix:



SpaceX Product Teams:



Each team owns the entire value stream from design through operations. All disciplines colocated, enabling spontaneous collaboration and eliminating handoff waste.

4.1.2 Decision Authority and Empowerment

SpaceX took decisions down to the competent minimum:

- Design, build, and test decisions are made by teams without the multi level approvals.
- Controls (safety requirements, interface specs, budget targets) are replaced by constraints.
- Design reviews of the CEO are done on a weekly basis thus making major decisions quickly.

Example: Raptor redesign that would cut 30 percent of parts and 50 percent cost of conventional aerospace would take 12-18 months of approvals; SpaceX team proposed to Musk approved at the same meeting implemented in 6 months.

4.2 Cultural Transformation

4.2.1 Embracing Failure: "Fail Fast, Fail Cheap, Fail in the Open"

Traditional: Failure is career-ending; blame culture; secrecy

SpaceX Inversion:

- Fail Fast: Prototype early with simple tests.
- Fail Cheap Commercial steel vs. aerospace titanium; Test articles that are disposable at $10-20M vs. a conventional test article of $100M+
- Public livestreams Fail: Explosions are educational.

After the SN8 landing explosion, Musk tweeted **"Mars, here we come!"** emphasizing successful flight portions, normalizing failure as progress.

4.2.2 Hiring for Mindset

SpaceX deliberately hires from **non-aerospace backgrounds**:

- Software engineers (iterative mindset)
- Automotive engineers (rapid prototyping experience)
- Video game developers (real-time systems comfort)
- Recent graduates (not indoctrinated)

Musk asked candidates: **"Why work on rockets?"** Rejected "career advancement," hired to "make humanity multi-planetary." Compromise: 20-30% reduced compensation, 60-80 hour weeks, 3-5 years on average but engineers will undertake on behalf of impact and excitement.

5.1 Quantitative Outcomes

Development Speed: 2.6x faster than traditional aerospace

SpaceX has shown faster development cycles by far than conventional aerospace ones. The Falcon 9 was built over a time of about 4.5 years at a cost of $300 million which was validated by NASA (NASA, 2011). Crew Dragon is a NASA Commercial Crew contract that was created in 8 years and provided a price that was 2.6 billion US dollars (NASA, 2014).

System	Organization	Development Time	Cost
Falcon 9	SpaceX	4.5 years	$300M
Crew Dragon	SpaceX	8 years	$2.6B
SpaceX Average		**5 years**	
SLS	NASA/Boeing	15+ years	$25B+
Starliner	Boeing	10+ years	$5.7B
Traditional Average		**13 years**	

Compare this to NASA's Space Launch System that has been under development since 2011, having cost 23.8 billion in the first flight (The Planetary Society, 2022), and the GAO has also found the program to be unaffordable at its present cost levels (GAO, 2023). The Boeing Starliner that was initially a contracted aircraft at 4.2 billion dollars has gone beyond its budget of more than 2 billion dollars as of 2025 (Boeing Starliner, 2025).

Cost Reduction: 90%+ decrease in launch costs

The spaceflight transformation in launch economics is one of the most important accomplishments in the history of spaceflight. The Space Shuttle of NASA was estimated to launch 27,500 kg to Low Earth Orbit at a cost of 1.5 billion dollars which is 54500 per kilogram (Jones, 2020). The Falcon 9 of SpaceX has a mass of 22,800 kg to the LEO at 62-67 million, about 2,720 per kilogram, which is 20 times cheaper (Jones, 2020).

System	Cost per kg to orbit
Space Shuttle	$54,500
Traditional (Atlas, Delta)	$10,000-$14,000
Falcon 9 (reused)	$1,500
Improvement	97% reduction

This cost revolution has "substantially reduced the cost of space launch" and created "greatly expanded opportunities to exploit space" for NASA and commercial customers alike (Jones, 2020).

Deployment Frequency: 10-15x improvement

The pace of launch at SpaceX has changed the industry norms. In 2023, SpaceX conducted 96 Falcon family launches (91 Falcon 9 and 5 Falcon Heavy), surpassing both the company's own single-year record of 61 and the global annual record of 64 launches previously held by the Soviet R-7 family (Wikipedia, 2025).This constituted 43 percent of all Launches in the world than any individual country (NASASpaceFlight, 2023).

- **Falcon 9 (2023):** 96 launches (CNBC, 2023)
- **Falcon 9 (2024):** 134 launches (Wikipedia, 2025)
- **Traditional systems:** 1-2 per year

The company achieved pad turnaround times of under 4 days at Cape Canaveral's SLC-40, with goals to reduce this to under 24 hours (NASASpaceFlight, 2023).

Quality/Reliability: Learning curve advantage

Even with such a fast innovation strategy, SpaceX has reached a superb level of reliability. Falcon 9 family has a 99.49 success rate, having already conducted 584 launches and with the latest Block 5 model recording a 99.81 success rate in 516 missions (Wikipedia, 2025).

- **Falcon 9 success rate:** 99%+ (Wikipedia, 2025)
- **Mean Time to Recovery:** 4-6 months vs. traditional 12-24 months
- **Booster reuse record:** 31 flights on a single booster (Wikipedia, 2025)

This shows that fast iteration, with intensive testing and learning systems, yields more reliability in the final result than conservative development strategies.

5.2 Market Impact

SpaceX Market Share: 0% (2010) - 50%+ (2024) of global orbital launch

Competitive Response (industry-wide transformation):

- **Arianespace:** Abandoned Ariane 5, developing Ariane 6 (reusable)
- **ULA:** Vulcan rocket with cost reduction focus
- **Blue Origin:** New Glenn incorporates reusability
- **China:** Long March 8/9 adding reusability

6. Critical Analysis and Lessons

6.1 Key Success Factors

The use of the eight-step change process as proposed by Kotter (1996): SpaceX has generated urgency (commercial opportunity + aerospace stagnation), guiding coalition (Musk + technical leadership), developed vision (Mars colonization), communicated vision (public testing), empowered action (eliminated bureaucracy), generated short-term wins (Falcon 1 success), consolidated gains (expanded to Dragon/Starship) and anchored change (cultural mechanisms).

6.2 Replicability and Boundaries

Favourable Conditions (Applicable Agile hardware): Technology maturity (fundamentals learned-Christensen, 1997); acceptable prototype costs (Ries, 2011); acceptable failure implications (Beck et al., 2001); flexible regulations; market pull (Womack and Jones, 2003); patient capital.

Unfavorable Conditions (traditional superior): System of One-of-a-kind Unfavorable: failure results in mission loss (James Webb Telescope); cost of iteration is prohibitive (particle accelerators); zero tolerance (life-safety) Unfavorable: very inflexible regulations (nuclear energy); basic physics is still unknown.

The Boundary: Agile is applied to systems of production that have the iterative learning curve and several units are going to be constructed and the improvement of data feeds should be presented. The classical waterfall is suitable to individual systems when there is only one instance that has to be perfect and the failure situation is disastrous (Poppendieck and Poppendieck, 2003; Rigby et al., 2016).

6.3 Cross-Industry Applications

Where It Works: Automotive (Tesla has already done it--Vance, 2015); military contracts (Anduril, Palantir coming); consumer technology (Apple, Xiaomi has already implemented).

Where It Doesn't: Pharmaceuticals (FDA experimentation incompatible with high-speed iteration); commercial aviation (Boeing 737 MAX showed safety critical system risks of moving fast); nuclear energy (society has zero tolerance towards incidents).

6.5 Validation Boundaries and Ongoing Demonstration

The Agile approach that SpaceX has used in its Earth-orbital systems (with 250+ flights at 99%+ reliability) is still being tested with deep-space super-heavy-lift vehicles. Starship with a capability superior to that of Saturn V, chosen as NASA Artemis Human Landing System (2.9B contract, NASA 2021) has already undergone five test flights (as of late 2024) but has yet to demonstrate the full functionality (Henry, 2024).

NASA's Artemis program uses a hybrid experience of the Artemis program of NASA uses a combination of traditional and Agile approaches: a Space Launch System (SLS) designed in a waterfall method of about 25 billion dollars over 15 years and created (GAO, 2023) launches crews to lunar orbit, whereas Starship built in an iterative manner of approximately 3 billion dollars over five years will carry out lunar landings. This mixed paradigm implies that even more conservative organizations will admit that Agile is more cost-effective when it comes to iterative systems, and that waterfall methods will be used to develop flight-proven technology that needs minimal iteration. Starship's successful operational deployment will provide critical validation of whether Agile hardware development scales to humanity's most ambitious engineering challenges.

7. Conclusion

SpaceX demonstrated that the Agile, Lean, and DevOps can be useful to hardware, but not only to software. They reduced the time to develop the product by 2.6x and the cost by 90% to demonstrate that the conventional aerospace industry was not in a position to pursue the use of Cold War-era methods in the midst of Cold War-era budgets (Weinzierl et al., 2018; GAO, 2023).

The change was not one that involved a couple of Agile practices that it needed to adopt. The departmental silos were substituted with cross-functional teams. The engineers retained the ownership of their work all the way to flight. Failure in the public tests turned into learning opportunities, rather than career disasters. Most importantly, fixed-price contracts forced efficiency, unlike cost-plus deals that rewarded waste (Kotter, 1996; Denning, 2018).

These principles can work in other industries. Tesla did it with cars, Anduril with defense systems but only when the context fits. You need reasonable prototype costs, room to fail during development, and the authority to restructure organizations. It doesn't work for one-off projects like the James Webb Telescope or life-critical systems like pharmaceuticals where developmental failures aren't acceptable (Rigby et al., 2016).

What's fascinating is NASA's Artemis program using both approaches: traditional SLS ($25B, 15 years) for the launch, Agile-developed Starship ($3B, 5 years) for landing. Even NASA admits the old way costs too much. When Starship becomes fully operational, we'll see if Agile hardware development really scales to the hardest engineering problems humanity faces.

The bigger lesson: processes stick around not because they're best, but because they're familiar. Apollo's waterfall made sense when money was unlimited and failure meant losing to the Soviets. That world ended in 1991, but aerospace kept doing things the same way. SpaceX won by asking the right question not "how do we do this better?" but "if we started from scratch today, would we do it this way at all?" (Christensen, 1997). That's the real transformation.

References

Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., & Thomas, D. (2001). *Manifesto for Agile software development*. Agile Alliance. https://agilemanifesto.org

Berger, E. (2020). *Liftoff: Elon Musk and the desperate early days that launched SpaceX*. William Morrow.

Boeing Starliner. (2025, November 28). In *Wikipedia*. https://en.wikipedia.org/wiki/Boeing_Starliner

Christensen, C. M. (1997). *The innovator's dilemma: When new technologies cause great firms to fail*. Harvard Business School Press.

CNBC. (2023, December 29). *SpaceX sets new rocket record with 96 successful launches in 2023*. CNBC. https://www.cnbc.com/2023/12/29/spacex-rockets-2023-launch-record.html

Denning, S. (2018). *The age of Agile: How smart companies are transforming the way work gets done*. AMACOM.

Edmondson, A. (1999). Psychological safety and learning behavior in work teams. *Administrative Science Quarterly*, *44*(2), 350-383. https://doi.org/10.2307/2666999

Forsgren, N., Humble, J., & Kim, G. (2018). *Accelerate: The science of lean software and DevOps: Building and scaling high performing technology organizations*. IT Revolution Press.

Foust, J. (2024). Boeing records additional losses on Starliner program. *SpaceNews*. https://spacenews.com

Galbraith, J. R. (2009). *Designing matrix organizations that actually work: How IBM, Procter & Gamble, and others design for success*. Jossey-Bass.

Henry, C. (2024). SpaceX Starship development timeline and milestones. *SpaceNews*. https://spacenews.com

Jones, H. W. (2020). *The recent large reduction in space launch cost* (NASA/TM-2020-220001). NASA Technical Reports Server. https://ntrs.nasa.gov/citations/20200001093

Kim, G., Debois, P., Willis, J., & Humble, J. (2016). *The DevOps handbook: How to create world-class agility, reliability, and security in technology organizations*. IT Revolution Press.

Kotter, J. P. (1996). *Leading change*. Harvard Business School Press.

Kranz, G. (2000). *Failure is not an option: Mission control from Mercury to Apollo 13 and beyond*. Simon & Schuster.

Launius, R. D. (2003). Public opinion polls and perceptions of US human spaceflight. *Space Policy*, *19*(3), 163-175. https://doi.org/10.1016/S0265-9646(03)00039-0

Liker, J. K. (2004). *The Toyota Way: 14 management principles from the world's greatest manufacturer*. McGraw-Hill.

List of Falcon 9 and Falcon Heavy launches. (2025, December 4). In *Wikipedia*. https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Logsdon, J. M. (2010). *John F. Kennedy and the race to the moon*. Palgrave Macmillan.

NASA. (2011, August). *Falcon 9 launch vehicle NAFCOM cost estimates*. https://www.nasa.gov/wp-content/uploads/2015/01/586023main_8-3-11_NAFCOM.pdf

NASA. (2014, September 16). *NASA chooses American companies to transport U.S. astronauts to International Space Station* [Press release]. https://www.nasa.gov/press-release/nasa-chooses-american-companies-to-transport-us-astronauts-to-international-space

NASA. (2018). *NASA systems engineering handbook* (NASA/SP-2016-6105, Rev 2). National Aeronautics and Space Administration. https://www.nasa.gov/seh

NASA. (2020). *Apollo program budget appropriations*. NASA History Division. https://www.nasa.gov/history

NASA. (2021). *NASA selects SpaceX to develop lunar landers for Artemis missions* [Press Release 21-108]. National Aeronautics and Space Administration. https://www.nasa.gov/press-release

NASA Vision 2040 Study Team. (2018). *Vision 2040: A roadmap for integrated, multiscale modeling and simulation of materials and systems* (NASA/CR-2018-219771). National Aeronautics and Space Administration.

NASASpaceFlight. (2023, December 31). *SpaceX closes out record-breaking 2023, prepares for more records in 2024*. NASASpaceFlight.com. https://www.nasaspaceflight.com/2023/12/falcon-roundup/

Ohno, T. (1988). *Toyota Production System: Beyond large-scale production*. Productivity Press.

Payload Space. (2025, January 3). *2024 orbital launch attempts by country*. Payload Space. https://payloadspace.com/2024-orbital-launch-attempts-by-country/

Pomeroy, T. (2025, January 15). *Space trends in 2024*. American Enterprise Institute. https://www.aei.org/op-eds/space-trends-in-2024/

Poppendieck, M., & Poppendieck, T. (2003). *Lean software development: An Agile toolkit*. Addison-Wesley Professional.

Ries, E. (2011). *The lean startup: How today's entrepreneurs use continuous innovation to create radically successful businesses*. Crown Business.

Rigby, D. K., Sutherland, J., & Takeuchi, H. (2016). Embracing Agile. *Harvard Business Review*, *94*(5), 40-50.

Rother, M., & Shook, J. (2003). *Learning to see: Value stream mapping to add value and eliminate muda*. Lean Enterprise Institute.

Schwaber, K., & Sutherland, J. (2020). *The Scrum guide: The definitive guide to Scrum: The rules of the game*. Scrum.org. https://scrumguides.org

Sheetz, M. (2023, December 29). SpaceX broke records in 2023 for orbital launches. *CNBC*. https://www.cnbc.com/2023/12/29/spacex-rockets-2023-launch-record.html

Space launch market competition. (2025, November). In *Wikipedia*. https://en.wikipedia.org/wiki/Space_launch_market_competition

The Planetary Society. (2022, August 2). *The cost of SLS and Orion*. https://www.planetary.org/space-policy/cost-of-sls-and-orion

U.S. Government Accountability Office. (2023). *NASA: Assessments of major projects* (Report No. GAO-23-106021). U.S. Government Accountability Office. https://www.gao.gov/products/gao-23-106021

U.S. Government Accountability Office. (2023). *Space Launch System: Cost transparency needed to monitor program affordability* (Report No. GAO-23-105609). U.S. Government Accountability Office. https://www.gao.gov/products/gao-23-105609

Vance, A. (2015). *Elon Musk: Tesla, SpaceX, and the quest for a fantastic future*. Ecco.

Weinzierl, M., Rousseau, B., & Goodwin, J. (2018). The commercial space age is here. *Harvard Business Review*, *96*(2), 92-100.

Weinzierl, M., & Sarang, M. (2020). SpaceX, economies of scale, and a revolution in space access (Harvard Business School Case 720-027). Harvard Business Publishing.

Womack, J. P., & Jones, D. T. (2003). *Lean thinking: Banish waste and create wealth in your corporation* (2nd ed.). Free Press.